

PROBLEM SHEET 2

Alex Kavvos

The following questions are about the language of numbers and strings.

1. Write down the abstract syntax tree for the pre-term `plus(let(len( $x$ );  $i$ . plus( $i$ ;  $n$ )); num[2])`.
2. Assume $\Sigma \stackrel{\text{def}}{=} \{0, 1\}$. Write a program that
 - has a free variable x of type `Str`,
 - appends the string `0110` to x ,
 - computes the length of the compound string, and
 - adds that number to itself.

Your program should not mention the string literal `str[0110]` more than once.

3. Produce a typing derivation for the following terms, assuming that $\Sigma \stackrel{\text{def}}{=} \{0, 1\}$.

(i) $x : \text{Str} \vdash x : \text{Str}$

(ii) $\vdash \text{plus}(\text{num}[1]; \text{num}[1]) : \text{Num}$

(iii) $x : \text{Str} \vdash \text{cat}(x; \text{str}[01]) : \text{Str}$

(iv) $x : \text{Str}, n : \text{Num} \vdash \text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2]) : \text{Num}$

4. Perform the following substitutions, step-by-step.

(i) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[i/x]$

(ii) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[\text{num}[0]/n]$

(iii) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[i/n]$

5. State the cases of the inversion lemma for the following constructs:

(i) `len( $e$ )`

(ii) `let( $e_1$ ;  $x$ .  $e_2$ )`

6. Prove the weakening lemma for the programming language of numbers and strings.

7. (*) Complete the proof of substitution from Lecture 4.

8. Prove that types are unique, i.e. that for every context Γ and pre-term e there exists at most one τ such that $\Gamma \vdash e : \tau$.

[Hint: assume that there exist two, and prove that they must be the same.]